pathogen

Purpose

The pathogen module simulates user-listed pathogens across soil, foliage, channel water, benthic sediment, and aquifers. The set of pathogens simulated is declared in constituents.cs, which populates `cs_db%num_paths`. The files below load the parameter database and the per-object initial conditions.

Activation

The pathogen database `pathogens.pth` is referenced from `file.cio` as `in_parmdb%pathcom_db` (default `pathogens.pth`). The initial-condition files are referenced via `in_init%path_soil` (default `path_hru.ini`) and `in_init%path_water` (default `path_water.ini`).

`path_parm_read` is called from `proc_db.f90`. `path_hru_aqu_read` is called from `proc_read.f90`. `path_cha_res_read` is called later from `main.f90`.

Files

pathogens.pth

Pathogen parameter database. One row per pathogen.

- Reader: src/path_parm_read.f90
- Type: `path_db(0:path)` (`type pathogen_db`) in src/pathogen_data_module.f90.
- Format: title line, header line, then one line per pathogen.

#	Field	Units	Description
1	pathnm	-	pathogen name
2	do_soln	1/day	die-off factor in soil solution
3	gr_soln	1/day	growth factor in soil solution
4	do_sorb	1/day	die-off factor sorbed to soil particles
5	gr_sorb	1/day	growth factor sorbed to soil particles
6	kd	none	partition coefficient between solution and sorbed phase in surface runoff
7	t_adj	none	temperature adjustment factor for die-off and growth
8	washoff	none	fraction on foliage washed off by a rainfall event
9	do_plnt	1/day	die-off factor on foliage
10	gr_plnt	1/day	growth factor on foliage
11	fr_manure	none	fraction of manure with active colony forming units
12	perco	none	percolation coefficient, ratio of solution bacteria in the surface layer
13	det_thrshd	cfu/m^2	detection threshold below which soil pathogen levels are zeroed
14	do_stream	1/day	die-off factor in streams
15	gr_stream	1/day	growth factor in streams
16	do_res	1/day	die-off factor in reservoirs
17	gr_res	1/day	growth factor in reservoirs
18	conc_min	-	minimum pathogen concentration

path_hru.ini

Initial pathogen concentrations for HRU soils and plants. One block per record.

- Reader: src/path_hru_aqu_read.f90
- Type: `path_soil_ini` (`type cs_soil_init_concentrations`) in src/constituent_mass_module.f90. Fields: `name`, `soil(num_paths)`, `plt(num_paths)` in cfu/m^2.
- Format: one title line, then per record: a header line, a name line, then one line containing `name, soil(:), plt(:)` for all pathogens in `constituents.cs` order.

The same reader allocates `cs_path_solsor(num_paths)`.

path_water.ini

Initial pathogen concentrations for channels and reservoirs (water column and benthic).

- Reader: src/path_cha_res_read.f90
- Type: `path_water_ini` (`type cs_water_init_concentrations`) in `src/constituent_mass_module.f90`. Fields: `name`, `water(num_paths)`, `benthic(num_paths)`. Record names are stored in `path_init_name`.
- Format: title line, header line, then per record: name line, then `num_paths` pairs of lines (water value, benthic value). On the second pass the reader reads a single line `name, water(:), benthic(:)` per record.

Related

- constituents.cs. Declares the pathogen list and sets `cs_db%num_paths`.
- pesticide module. Same per-object initial-condition pattern.
- metals module. Same per-object initial-condition pattern.